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Project Title :	Dynamic Portfolio Risk Arbitrage & Value-at-Risk (VaR) Monte Carlo Simulator
Week :	Week 5

Task: WEEK-5 – Business Analytics

1. Objective

Design a portfolio analytics system that studies how a collection of financial assets may behave under different market conditions. The system focuses on measuring portfolio uncertainty, generating possible future outcomes, and estimating downside risk.

- Process historical price information for multiple assets.
- Calculate periodic asset returns.
- Measure relationships between different assets.
- Generate correlated market scenarios.
- Simulate future portfolio values.
- Estimate potential portfolio losses.

2. Input Data

The system requires historical financial information for multiple assets.

Required inputs:

- Historical asset prices
- Portfolio allocation percentages
- Current asset prices
- Selected historical period
- Number of simulations
- Confidence level

3. Return Calculation

Historical prices are converted into returns to understand the movement of each asset.

Formula:

Return = (Current Price − Previous Price) / Previous Price

For example, if an asset price increases from ₹200 to ₹210:

Return = (210 − 200) / 200 = 0.05 = 5%

Returns are used to calculate volatility and relationships between assets.

4. Portfolio Statistics

Calculate important statistics for each asset, such as:

- Average return
- Standard deviation
- Minimum return
- Maximum return
- Volatility

These measurements help understand the performance and risk level of individual assets.

5. Correlation and Covariance Analysis

A covariance matrix is used to determine how asset returns move together.

- Diagonal values represent the variance of individual assets.
- Off-diagonal values represent the relationship between pairs of assets.
- A correlation matrix can be used to visualize strong and weak relationships.

6. Cholesky Decomposition

The covariance matrix can be factorized using Cholesky decomposition.

$\boldsymbol{\Sigma = LL^T}$

Where:

- $\boldsymbol{\Sigma}$ = covariance matrix
- $\mathbf{L}$ = lower triangular matrix
- $\mathbf{L^T}$ = transpose of the matrix

This method helps generate random market movements while maintaining the relationships between assets.

7. Scenario Generation

Independent random values are generated and transformed using the Cholesky matrix.

Process:

Independent Random Values
↓
Cholesky Transformation
↓
Correlated Market Shocks
↓
Simulated Market Scenarios

Repeated scenarios provide different possible market conditions.

8. Geometric Brownian Motion

Future asset prices can be estimated using the Geometric Brownian Motion model.

The model considers:

- Current asset price
- Expected return
- Volatility
- Time interval
- Random market shock

For multiple assets, correlated random shocks can be used to make the simulation more realistic.

9. Monte Carlo Simulation

The model is executed thousands of times to create different possible portfolio outcomes.

Each simulation represents one possible future market condition.

The results form a **portfolio-value distribution**, showing possible gains and losses.

10. Portfolio Profit and Loss

For every simulation, calculate the portfolio value using the asset weights.

Portfolio Value = Σ (Portfolio Weight × Simulated Asset Price)

Then calculate:

Profit/Loss = Future Portfolio Value − Initial Portfolio Value

This helps identify potential positive and negative outcomes.

11. 99% Value-at-Risk

99% VaR is used to estimate the potential downside risk of the portfolio.

For example:

Initial Portfolio Value = ₹10,00,000
99% VaR = ₹1,00,000

This means the model estimates a loss threshold of approximately ₹1,00,000 in the worst 1% of simulated outcomes, based on the model assumptions.

VaR depends on:

- Confidence level
- Time horizon
- Historical data
- Volatility
- Asset correlations
- Simulation method

12. Overall System Flow

Historical Asset Prices
↓
Data Cleaning
↓
Calculate Returns
↓
Calculate Volatility & Correlation
↓
Create Covariance Matrix
↓
Cholesky Decomposition
↓
Generate Random Market Shocks
↓
Create Correlated Scenarios
↓
Simulate Future Prices
↓
Calculate Portfolio Values
↓
Calculate Profit/Loss
↓
Calculate 99% VaR
↓
Risk Analysis Dashboard

13. Expected Output

The system should produce:

1. Asset return statistics
2. Volatility measurements

3. Covariance matrix
4. Correlation matrix
5. Cholesky factor
6. Simulated asset prices
7. Portfolio-value distribution
8. Profit/loss distribution
9. 99% VaR
10. Risk interpretation
11. Optional Expected Shortfall (CVaR)
12. Risk visualization charts

14. Technologies You Can Use

Programming Language: Python

Libraries:

- **NumPy** – numerical and matrix calculations
- **Pandas** – data processing
- **SciPy** – statistical calculations and decomposition
- **Matplotlib** – data visualization
- **Seaborn** – correlation and statistical visualization

15. Key Learning Outcomes

After completing this task, students will understand:

- Portfolio risk measurement
- Financial data processing
- Return and volatility calculation
- Covariance and correlation
- Matrix operations
- Cholesky decomposition
- Monte Carlo simulation
- Geometric Brownian Motion
- Value-at-Risk
- Python-based financial analytics